KAIZEN MATHS

Lesson starter

Algebraic substitution

Expression substitution - mixed practice

4 questions · about 8 minutes

Name _____

Substitute, then expand and simplify.

S1 Given $x = 5y - 5$

Write in terms of y and simplify

$5x + 5$

S2 Given $a = 6b - 6$

Write in terms of b and simplify

$4a + 2b$

S3 Given $x = 3y + 5$

Write in terms of y and simplify

$3x$

S4 Given $a = 5b + 6$

Write in terms of b and simplify

$5a + 4b$

Show your working

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AS-11 · S · Starter

Show your working

KAIZEN MATHS

Exit ticket

Algebraic substitution

Expression substitution - mixed practice

Name _____

Simplify each expression.

E1 Given $x = 3y + 5$

Write in terms of y and simplify

$3x$

E2 Given $a = 5b + 6$

Write in terms of b and simplify

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AS-11 · E · 2 questions

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AS-11 · E · 2 questions

Show your working

Teacher answers & working

AS-11 · Expression substitution - mixed practice

Page 1: two starters. Page 2: four exit tickets. Cut along the dashed lines.

Print student pages single-sided at 100%. Match S and E labels to the solutions below.

S1

Given x=5y-5

$5x + 5$

Write in terms of y and simplify

Substitute x=5y-5.

$5(5y - 5) + 5$

$25y - 25 + 5$

$25y - 20$

S2

Given a=6b-6

$4a + 2b$

Write in terms of b and simplify

Substitute a=6b-6.

$4(6b - 6) + 2b$

$24b - 24 + 2b$

$(24 + 2)b - 24 = 26b - 24$

S3 / E1

Given x=3y+5

$3x$

Write in terms of y and simplify

Substitute x=3y+5.

$3(3y + 5)$

$9y + 15$

$9y + 15$

S4 / E2

Given a=5b+6

$5a + 4b$

Write in terms of b and simplify

Substitute a=5b+6.

$5(5b + 6) + 4b$

$25b + 30 + 4b$

$(25 + 4)b + 30 = 29b + 30$